1 Introduction

This laboratory session focuses on understanding the control of a DC motor using the L298N dual H-Bridge motor driver, which allows for precise control over speed and direction. This session aims to explore both manual control of motor direction and speed modulation via Pulse Width Modulation (PWM) using an Arduino.

2 Experiments

2.1 Experiment 1: Testing the Motor Driver

In the first part of our experiments, the L298N motor driver was connected as specified in the Figure 1. The motor driver setup was configured to first control motor direction manually, and then with further modifications, using the Arduino for more refined control.

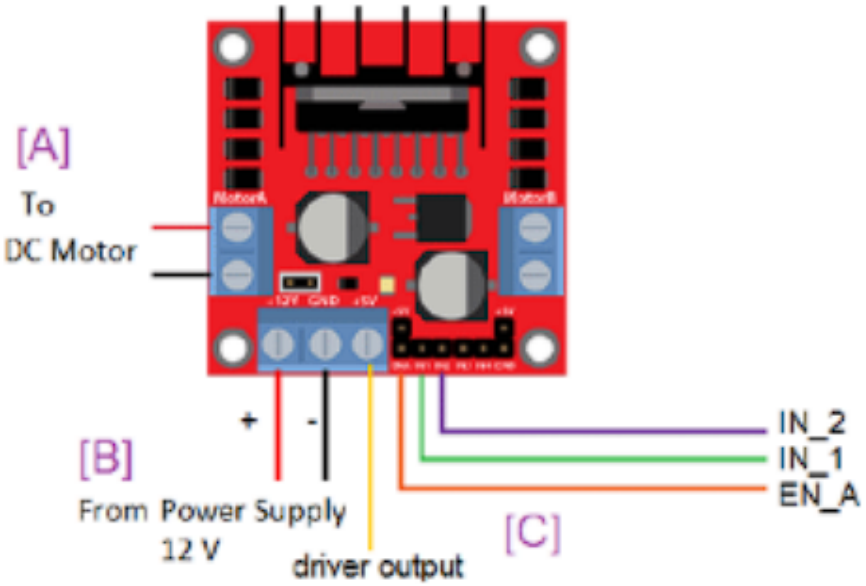


Figure 1: Connection L298N motor driver and Arduino.

2.1.1 Components Used:

- Arduino Uno
- L298N Motor Driver
- DC motor with encoder
- Power supply
- Oscilloscope
- Jumper cables
- Breadboard

2.1.2 Setup and Observations:

The initial setup involved connecting the motor driver to the DC motor and Arduino. The EN-A pin was connected to a digital pin on the Arduino to control speed. This setup aimed to observe the effects of different voltage levels on motor operation through PWM provided by the Arduino.

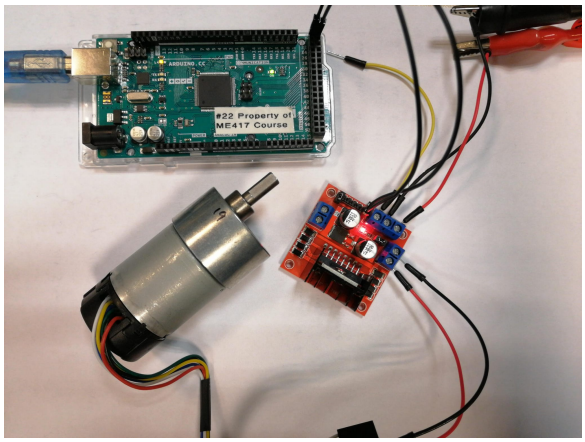


Figure 2: Initial setup with L298N motor driver and Arduino

Question A: What is the difference between giving high voltage to IN1 and IN2 in terms of motor rotation direction?

Answer: Giving high voltage to IN1 causes the motor to rotate in one direction, while giving it to IN2 causes it to rotate in the opposite direction. This is due to the way the H-Bridge in the L298N driver controls the motor's direction by reversing the current flow through the motor.

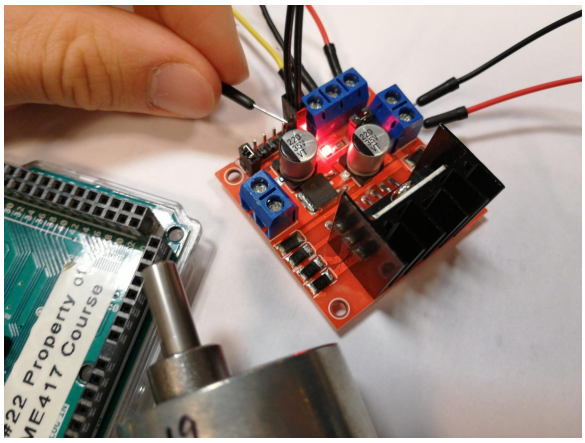


Figure 3: Changing input IN1 to IN2 on L298N motor driver.

2.2 Experiment 2: Controlling Motor Speed with Arduino

For this experiment, the setup from Experiment 1 was modified to enable PWM control over the motor speed. The Arduino function `analogWrite()` was utilized to adjust the motor speed dynamically within the experiment's constraints.

2.2.1 Arduino Code Implementation:

```
#define enA 2
#define in1 3
#define in2 4

void setup() {
  Serial.begin(9600);
  pinMode(enA, OUTPUT);
  pinMode(in1, OUTPUT);
  pinMode(in2, OUTPUT);
}

void loop() {
  analogWrite(enA, 150); // Setting motor speed
  digitalWrite(in1, HIGH);
  digitalWrite(in2, LOW);
  delay(2000); // Motor runs for 2 seconds in one direction
  digitalWrite(in1, LOW);
  digitalWrite(in2, HIGH);
  delay(2000); // Motor runs for 2 seconds in the opposite direction
}
```

Question B: Which value will make the motor rotate at its maximum speed and which one for the lowest speed?

Answer: The maximum speed of the motor is achieved by setting the PWM value to 255, and the lowest speed (but not zero) is achieved by setting it to a value close to but greater than 0. Setting it to 0 will completely stop the motor.

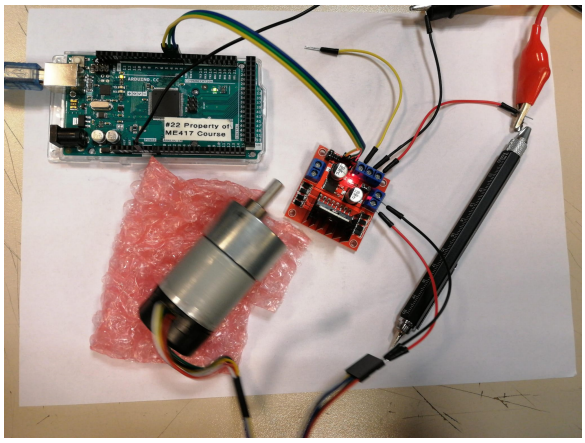


Figure 4: Controlling motor speed using PWM

3 Conclusion

The experiments demonstrated the capabilities of the L298N motor driver in controlling a DC motor's speed and direction. Using the Arduino, we successfully modulated the PWM signal to adjust the motor speed and direction. This laboratory session provided practical experience with motor drivers and highlighted the importance of electronic control systems.